GMI German-Malaysian Institute

DIPLOMA PROGRAMME

THEORY TEST 1

Academic Period: July 2026

Subject	:	**INTERNET OF THINGS**
Code	:	**CBS 2383**
Duration	:	**1 HOUR**
Total Marks	:	**50**

GENERAL INSTRUCTIONS:

1. Write down clearly (with capital letters) your name, trade, group and the date of examination at the bottom of this page.
2. Do not open this booklet until you are told to do so by the invigilator.
3. Read the specific instructions and s carefully before answering.

This paper consists of **7** printed pages including this page.

Student's Name	:	ABDUL HARIZ BIN MOHD SARNUDIN
Student's I/D No.	:	CBS24070208
Semester/Trade/Group	:	SEM 5 / CBS / DCBS 1
Date	:	15th September 2026

Raw Marks:	50
Percentage (in Doc 01):	15

No		Full Marks	Student's Marks	CLO	Level of Difficulty	DK & DP (ETAC Programme Only)	PLO GMI PLO CBS 9 PLO - Start Intake July 2023	PLO GMI PLO CBS 11 PLO - Start Intake July 2026
Part I	1-2	2		1	3		3	2
	3-4	2		1	3		3	2
	5-6	2		1	3		3	2
	7-8	2		1	3		3	2
	9-10	2		1	3		3	2
	11-12	2		1	3		3	2
	13-14	2		1	3		3	2
	15-16	2		1	3		3	2
	17-18	2		1	3		3	2
	19-20	2		1	3		3	2
Part II	1-6	30		1	3		3	2
Part III	1-7						#N/A	#N/A
Part IV	1-5						#N/A	#N/A
Total Full Marks		**50**		**Total Student's Marks**		**0**		

SUMMARY

Note:
FM - Full Marks,
StM - Student's Mark

CLO	Raw Marks FM	Raw Marks StM	Percentage FM	Percentage StM	PLO GMI PLO CBS 9 PLO - Start Intake July 2023 PLO	Raw Marks FM	Raw Marks StM	Percentage FM	Percentage StM	PLO GMI PLO CBS 11 PLO - Start Intake July 2026 PLO	Raw Marks FM	Raw Marks StM	Percentage FM	Percentage StM
Total														
CLO 1	50		15.0		PLO 3	50		15.0		PLO 2	50		15.0	
CLO 2	0		0.0		PLO 2	0		0.0		PLO 3	0		0.0	
CLO 3	0		0.0		PLO 4	0		0.0		PLO 5	0		0.0	
CLO 4	0		0.0		#N/A	0		0.0		#N/A	0		0.0	
CLO 5	0		0.0		#N/A	0		0.0		#N/A	0		0.0	
CLO 6	0		0.0		#N/A	0		0.0		#N/A	0		0.0	
Total Marks	**50**		**15.0**											

Signature	( )

Part I: Answer all s. (20 Marks)

1. Which of the following is the most accurate definition of a sensor in an IoT system?

A. A device that converts electrical energy into physical mechanical movement.
B. A device that measures a physical property by detecting information from the physical world.
C. A controller that provides internet connectivity and acts as the brains of the system.
D. A software program that processes stored data on a remote cloud server.

2. What is the primary function of an actuator in a controlled IoT system?

A. To collect environmental data and transmit it to a central controller.
B. To uniquely identify devices on a network using physical and logical addresses.
C. To transform an electrical signal from a controller into a physical output.
D. To regulate voltage drops across a solderless prototyping board.

3. True or False: An open-loop control system uses a sensor to continuously adjust its input.

A. True
B. False
C. None
D. None

4. True or False: Standard layered s like OSI and TCP/IP assist in protocol design and provide a common language to describe network functions.

A. True
B. False
C. None
D. None

5. According to Ohm's Law, what is the formula to calculate Voltage (V)?

A. V = I / R
B. V = R / I
C. V = I * R
D. V = I + R

6. If an electrical component has a voltage (V) of 12V and a current (I) of 3A, what is its resistance (R) based on Ohm's Law?

A. 36 ohms
B. 4 ohms
C. 0.25 ohms
D. 9 ohms

7. Which formula should be used to calculate Current (I) in a circuit when Voltage (V) and Resistance (R) are known?

A. I = V * R
B. I = R / V
C. I = V / R
D. I = V - R

8. What is the standardized unit of measurement for electrical resistance?

A. Amperes (amps)
B. Volts (V)
C. Watts (W)
D. Ohms (Ω)

9. What is the primary program development environment used to write and upload instructions to an Arduino microcontroller?

A. Python Shell CLI
B. Arduino Integrated Development Environment (IDE)
C. Raspbian Terminal Emulator
D. Jupyter Notebook with PL-App

10. What is the definition of computer code or a program?

A. An unorganized list of hardware specifications and wiring ports.
B. A set of ordered instructions created to accomplish a specific task.
C. An interactive diagram used to map custom business finances.
D. A physical connection cable that links sensors to controllers.

11. Which programming structure is used to execute a block of code repeatedly while a specific condition remains true?

A. IF-THEN statement
B. WHILE loop
C. FOR loop
D. Case-Select block

12. What tool allows users to access and program the Raspberry Pi remotely over a network without needing a local monitor, keyboard, or mouse?

A. Arduino IDE Serial Monitor
B. PL-App (Physical Laboratory Application)
C. Standard Packet Tracer Logical Workspace
D. Windows Device Manager

13. Raspbian, the operating system created specifically for the Raspberry Pi, is based on which Linux distribution?

A. Red Hat
B. Ubuntu
C. Debian
D. Slackware

14. What type of network usually spans a few meters around an individual and is commonly used to connect wearable IoT devices?

A. Wide Area Network (WAN)
B. Personal Area Network (PAN)
C. Local Area Network (LAN)
D. Low Power Wide Area Network (LPWAN)

15. What are the two specialized protocols used by resource-constrained embedded IoT devices to enable communication with low power and limited connectivity?

A. HTTP and FTP
B. CoAP and MQTT
C. SSH and Telnet
D. SMTP and POP3

16. Which wireless protocol is a low-energy, low-power, low-data rate technology used to create personal area networks for home automation and medical device data collection?

A. 4G LTE
B. ZigBee
C. Wi-Fi (802.11ac)
D. LoRaWAN

17. What is the term used to describe 'data about data' that is often embedded within a digital object and hidden from the user?
A. Routing table
B. Metadata
C. Firmware
D. System sketch

18. Which of the following is a physical factor that increases the difficulty of securing IoT devices?
A. The use of standardized application protocols like REST
B. The non-traditional physical location of devices
C. The high processing power available in embedded chips
D. The requirement of periodic automatic software updates

19. How does the Fog Computing differ from the Cloud Computing in IoT Architecture?
A. Fog computing stores all organizational data in large remote geographic data centers.
B. Fog computing positions distributed computing infrastructure closer to the network edge.
C. Fog computing replaces all physical microcontrollers with virtual software modules.
D. Fog computing operates entirely without internet connections or IP protocols.

20. What is the most common application of the 'Industrial Internet'?
A. Social media room-based collaboration
B. Predictive maintenance
C. Automated home appliance control
D. Wireless cellular routing mesh

ANSWER SHEET:

1.B	5.C	9.B	13.C	17.B
2.C	6.B	10.B	14.B	18.B
3.B	7.C	11.B	15.B	19.D
4.A	8.D	12.B	16.B	20.C

Part II: Answer all s. (30 Marks)

1: Explain the general concept of the Internet of Things (IoT) and how it benefits individuals and industries.

Answer:

The Internet of Things (IoT), is a network of physical objects embedded with sensors, software and network connectivity that allow it to collect and exchange data over the internet without requiring human intervention.

2: Describe the purpose of the Cisco IoT System and list its six foundational pillars.

Answer:

The purpose of the Cisco IoT System is to simplify the complexity of Internet of Things (IoT) deployments by providing a comprehensive, secure and scalable infrastructure. It bridges the gap between physical Operational Technology (OT) environments, such as manufacturing floors, utility grids and smart cities. And for traditional Information Technology (IT) networks, allowing organizations to connect, manage and secure billions of smart devices.

3: Explain the unique functions of sensors, actuators, and controllers in an IoT system.

Answer:

The unique functions of sensors, actuators and controllers are that they form the core input-process-output loop that allows an Internet of Things system to interact with the physical world.

4: Discuss physical connections in IoT and identify three common media types.

Answer:

The physical connections in IoT refer to the actual medium used to transmit data signals between devices, sensors, gateways and the internet itself. These connections form the Physical Layer (Layer 1) in the OSi model.

The three most commonly used media types for physical connections in IoT are **Copper Cabling**, either Twisted-Pair or Coaxial. It is an insulated copper wire that transmits data using electrical signals that is ideal for fixed Industrial IoT setups. Other than that, the most known cable are **Fiber-Optic Cabling**, which is an ultra-thin strands of glasses or plastics that transmits data as a pulses of light which make them lightning fast in transmitting data, commonly used for long distance backhaul connections, smart city infrastructure and critical traffic nodes. And lastly, **Radio Waves** which is a wireless medium. Data is modulated onto electromagnetic radio frequencies. The most dominant medium for consumer and distributed IoT. It connects everything from smart home thermostats and wearable fitness trackers to remote agricultural sensors.

5: Differentiate between Direct Current (DC) and Alternating Current (AC) and identify their generation sources.

Answer:

The difference between Direct Current and Alternating current is that Direct Current is an electrical current that flows continuously in a single constant direction with a frequency of zero. Where Alternating current periodically reverses its direction and changes its magnitude over time.

6: Explain why programmers are highly valued in the modern job market and identify four distinct domains they work in.

Answer:

Programmers are highly valued in the modern job market because they build, maintain and secure the digital infrastructure that powers the global economy. As business and organization across every industry undergo digital transformation, software has changed from a secondary support tool into the primary driver of revenue, automation and customer engagement.